

Tomalika Mazumdar

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SUMMARY

Electrical and Electronic Engineering graduate from Bangladesh University of Engineering and Technology (BUET) with a strong foundation in renewable energy systems, power-system analysis, optimization, embedded systems, and electrical engineering. Experienced in MATLAB/Simulink-based simulation, power-system studies, hardware implementation, and interdisciplinary engineering projects. Undergraduate thesis focused on multi-objective optimization of hybrid PV-Battery-Diesel systems using Enhanced Electromagnetic Field Optimization and the Epsilon-Constraint method. Interested in renewable energy, power-system optimization, smart grids, sustainable electrical systems, and emerging energy technologies.

EDUCATION

Bangladesh University of Engineering and Technology (BUET) Jan 2022 – Jun 2026

B.Sc. Engg. in Electrical and Electronic Engineering

- CGPA: 3.40/4.00

Holy Cross College Jun 2018 – Jan 2021

Higher Secondary Certificate (HSC)

- GPA: 5.00/5.00

Bangla Bazar Govt. Girls' High School Jan 2007 – May 2018

Secondary School Certificate (SSC)

- GPA: 5.00/5.00

RESEARCH EXPERIENCE

Undergraduate Thesis: Enhanced Electromagnetic Field Optimization with Epsilon-Constraint for Optimal Sizing of Hybrid PV-Battery-Diesel System in Sandwip Island, Bangladesh

Supervisor: Dr. Ahmed Zubair, Professor, Department of EEE, BUET

- Developed a multi-objective optimization framework for optimal sizing of hybrid PV-Battery-Diesel systems.
- Applied Enhanced Electromagnetic Field Optimization (EEFO) integrated with the Epsilon-Constraint method to obtain Pareto-optimal solutions.
- Evaluated system performance using Net Present Cost (NPC), Cost of Energy (COE), renewable energy penetration, and system reliability indices.
- Conducted a case study for Sandwip Island, Bangladesh, considering local renewable resources and load demand characteristics.
- Analyzed the economic, technical, and reliability performance of different hybrid energy-system configurations using simulation and optimization techniques.

PROFESSIONAL EXPERIENCE

Dhaka Electric Supply Company PLC (DESCO) Nov 2025

Industrial Attachment – Sales & Distribution Division, Bashundhara, Dhaka

- Participated in industrial attachment activities covering electricity distribution, metering, billing, and load management.
- Gained exposure to substation operation and maintenance, control-room activities, and distribution-line maintenance.
- Observed pre-paid metering, power-factor monitoring and upgrading, new connections, and load sanction and retention procedures.
- Studied disconnection and reconnection procedures and One Point Service Center operations.
- Gained exposure to wireless and telecommunication systems and Distribution Automation System (DAS) maintenance.

NOTABLE PROJECTS

Frequency and Voltage Control of a Microgrid using Particle Swarm Optimization

- Developed a microgrid control model using MATLAB/Simulink.
- Applied Particle Swarm Optimization (PSO) for frequency and voltage regulation.

Underfrequency Load Shedding Implementation

- Simulated underfrequency load shedding schemes for maintaining power-system stability.

- Evaluated system recovery and frequency response during disturbances.

Temperature Dependent VFD

- Designed a temperature-dependent variable frequency drive control mechanism.
- Implemented automatic control based on temperature variation.

Construction of a Shell Type Transformer of Certain 100 VA Rating

- Designed and constructed a shell-type transformer with specified rating and performance characteristics.
- Analyzed transformer efficiency and operational behavior under load conditions.

Obstacle Avoiding Drone

- Developed a drone system capable of autonomous obstacle detection and avoidance.
- Integrated sensor-based navigation techniques.

Digital Vending Machine FSM

- Designed a finite state machine (FSM) based vending machine using Verilog HDL.
- Simulated digital logic behavior and state transitions.

Ultrasonic Object Mapping Bot

- Developed a robotic mapping system using ultrasonic sensing.
- Implemented obstacle detection and environmental scanning functionality.

DC Voltmeter

- Designed and implemented a DC voltmeter circuit for voltage measurement applications.

Frequency Modulation-Based Image Transmission via Radio Wave

- Implemented image transmission using frequency modulation techniques over radio communication systems.

Brain Tumor Detection on MRI Image

- Applied image processing techniques for automated brain tumor detection using MRI images.

Solving Poisson's Equation of Electrostatics using Numerical Methods

- Solved two-dimensional electrostatic field problems using numerical techniques.
- Analyzed potential distribution and convergence behavior.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, MATLAB, Verilog HDL

Developer & Simulation Tools: MATLAB/Simulink, HOMER Pro, AutoCAD, LTspice, PSpice, PSAF 2.81

Electrical & Hardware: Electrical System Design, Control Systems, PCB Design, Hardware Implementation, Simulation and Analysis

Laboratory Equipment: Oscilloscope, Function Generator, Digital Multimeter

Soft Skills: Team Collaboration, Problem Solving, Technical Presentation, Project Coordination

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Treasurer, Murchona – Musical Club of BUET

2025 – 2026

- Managed budgeting and financial coordination for club activities and events.
- Coordinated organizational activities with executive members.

Assistant Office Secretary, Kantho – Recitation Club of BUET

2025 – 2026

- Assisted in administrative and communication-related activities.
- Supported event organization and documentation.

RESEARCH INTERESTS

Renewable Energy Systems, Solar PV and Hybrid Energy Systems, Power System Optimization, Smart Grid Technologies, Microgrids, Energy Storage Systems, Embedded Systems, Control Systems, Optimization Techniques

ADDITIONAL INFORMATION

Languages: Bengali (Native), English (Professional Working Proficiency)

Interests: Renewable Energy, Solar PV Systems, EV Technologies, Embedded Systems, Robotics, Technical Innovation, Cultural Activities